

Optimizing Neural Networks

Agenda

- Neural Networks Quiz
- Advanced Optimizers beyond SGD
- Dropout for Regularization
- Batch Normalization for Regularization
- Weight Initialization Techniques

Let's begin the discussion by answering a few questions on neural networks

Neural Networks Quiz

What does momentum in SGD with Momentum help achieve?

A

Accelerates convergence using past gradients

B

Adjusts learning rates adaptively

C

Reduces the variance of gradient updates

D

Prevents overshooting during optimization

Neural Networks Quiz

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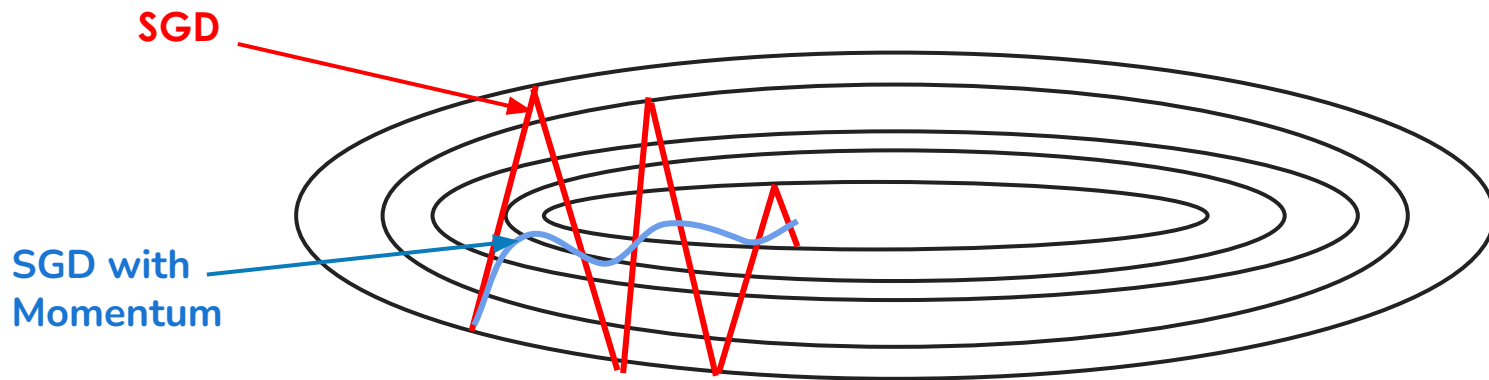
D

Prevents overshooting during optimization

Stochastic Gradient Descent with Momentum

SGD with momentum **accelerates convergence** by incorporating past gradients to maintain a consistent direction towards the minima

Momentum enhances this process by **reducing oscillations** and **facilitating smoother optimization trajectories**



Neural Networks Quiz

Which statement correctly describes how Adam and SGD with Momentum handle the learning rate?

A

The learning rate is constant throughout training for both Adam and SGD with Momentum.

B

Adam adjusts the learning rates separately for each parameter, while SGD with Momentum maintains a fixed learning rate.

C

SGD with Momentum changes the learning rate depending on gradient magnitude, whereas Adam sticks to a consistent learning rate.

D

Both Adam and SGD with Momentum utilize a dynamic learning rate approach throughout the training process.

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Learning Rates in ADAM & SGD with momentum

SGD with Momentum

Stochastic Gradient Descent with Momentum

Adds a momentum component

Fixed Learning Rate

Adam

Adaptive Moment Estimation

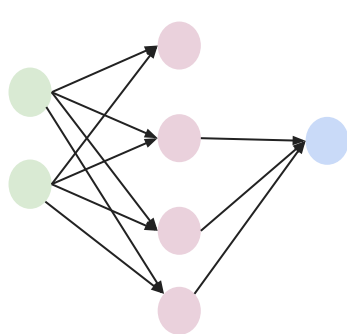
Adds a momentum component

Adaptive Learning Rate

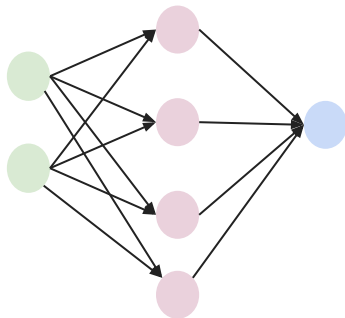
The learning rate is different for each model parameter and depends on the value of the gradient

Neural Networks Quiz

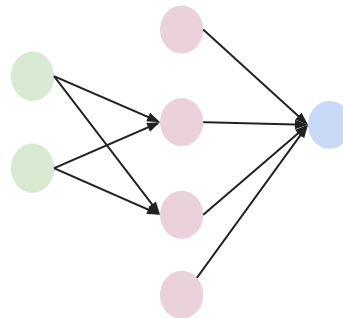
Which of the following neural networks best represents a dropout rate of 0.5?



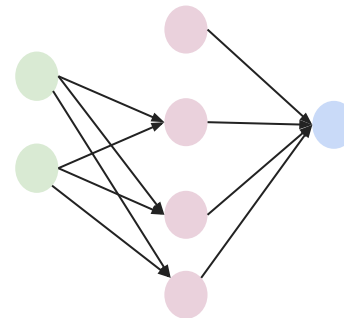
A



B



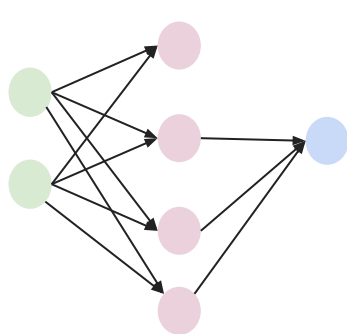
C



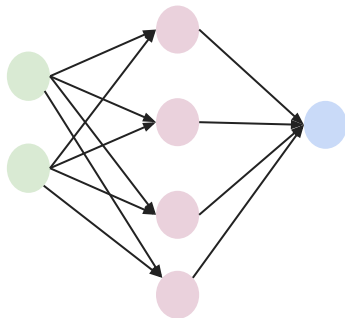
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Neural Networks Quiz

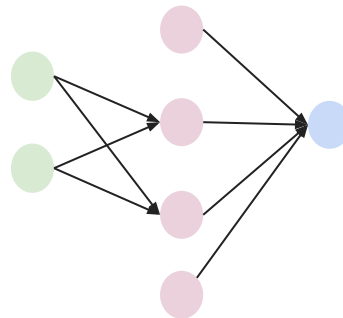
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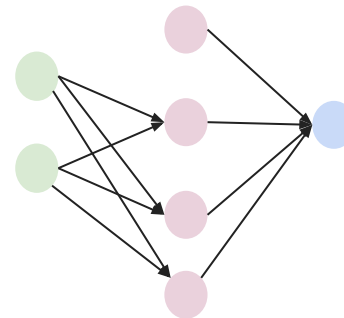
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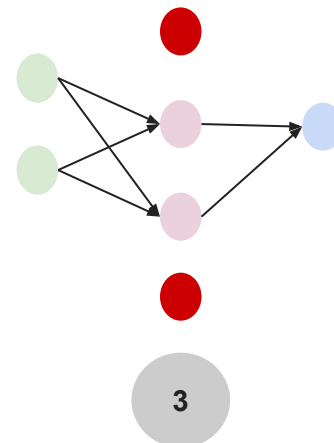
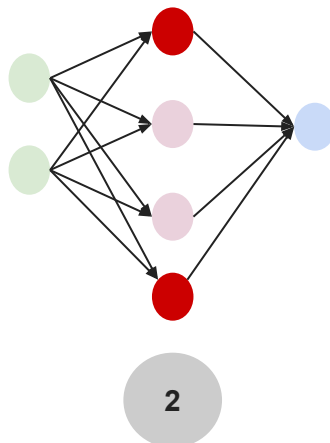
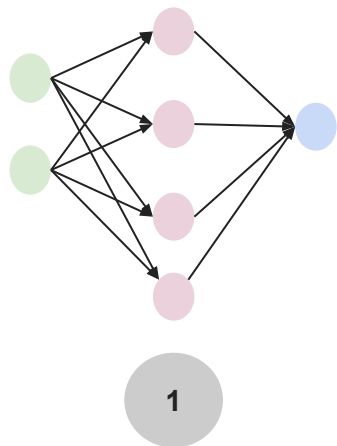
D

Dropout

Step 1 - Choose a dropout rate p ($p = 0.5$ here)

Step 2 - Randomly select $100 \cdot p\%$ of the neurons (50% here)

Step 3 - Deactivate the selected neurons by setting them to zero.



Neural Networks Quiz

How does Dropout contribute to dealing with overfitting in neural networks?

A

Decreasing the complexity of the neural network

B

Increasing the complexity of the neural network

C

Changing the activation function of a neuron

D

By adding Gaussian noise to the input

Neural Networks Quiz

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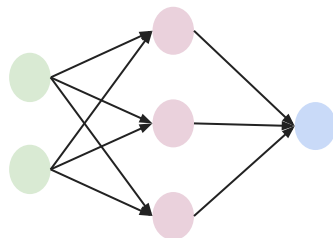
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Dropout for Regularization

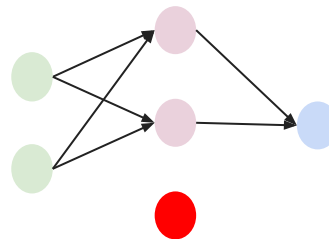
Dropout prevents overfitting by randomly deactivating neurons during training

Independence: Fosters independent learning among neurons

Ensemble: Creates a diverse ensemble of networks as neurons are deactivated randomly during training and the final result is an average prediction



Before Dropout



After Dropout

Neural Networks Quiz

How many learnable parameters are there in a batch normalization layer?

A

4

B

3

C

0

D

2

Neural Networks Quiz

How many learnable parameters are there in a batch normalization layer?

A

4

B

3

C

0

D

2

Batch Normalization - Working

Step 1 - Normalization: Normalize X (input) by subtracting its mean and dividing by its standard deviation.

No learnable parameters in Step 1

Step 2 - Scaling: Scale the output of Step 1 by multiplying it with a learnable parameter *gamma*

Step 3 - Shifting: Shift the output of Step 2 by adding an offset (learnable parameter *beta*)

Neural Networks Quiz

Which of the following accurately describes the purpose of batch normalization in neural networks?

A

Minimizing overfitting by adding noise to the input data.

B

Preventing vanishing gradients by initializing weights appropriately.

C

Reducing internal covariate shift by normalizing layer activations.

D

Decreasing model complexity by adding more parameters.

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Batch Normalization - Purpose

Stabilizing Training: Reduces internal covariate shift.

Regularization: Acts as a form of regularization.

Improved Gradient Flow: Enhances gradient flow for faster convergence.

Neural Networks Quiz

How does weight initialization contribute to improved learning in a neural network?

A

By introducing uniform gradients.

B

By ensuring symmetric neuron behavior.

C

By adjusting initial weights.

D

By hindering adaptation.

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Weight Initialization

Sets up the starting point for learning in a neural network

Helps break symmetry among neurons

Prevents gradient problems

Makes learning faster and more efficient overall

Neural Networks Quiz

What is the common problem that can arise when weights are initialized randomly in a neural network?

A

Vanishing Gradients

B

Exploding Gradients

C

Vanishing or Exploding Gradients

D

Learning rate becoming too high

Neural Networks Quiz

What is the common problem that can arise when weights are initialized randomly in a neural network?

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Vanishing or Exploding Gradients

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Weight Initialization

Randomly initializing weights in a neural network can lead to **two critical problems** during training

Vanishing gradients: Occurs when the gradients (derivatives) become extremely small, slowing down or halting the learning process

Exploding gradients: Occurs when the gradients become excessively large, causing instability and hindering effective learning

Specific weight initialization strategies help overcome these problems

Xavier (or Glorot) Initialization - sigmoid and tanh activations

He Initialization - ReLU and variants of ReLU



Happy Learning !

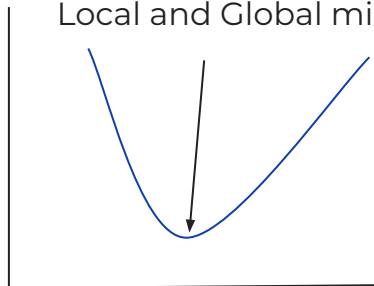


APPENDIX

Need for Advanced Optimizers

Convex Optimization

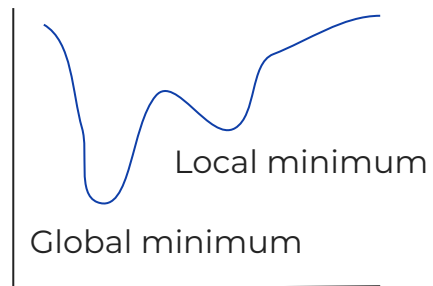
Local and Global minima



**Efficient optimization,
Global convergence**

**Guaranteed global
convergence under
conditions**

Non-Convex Optimization



**Complex, Local optima,
Challenging optimization**

**No guarantee, Prone to local
optima**